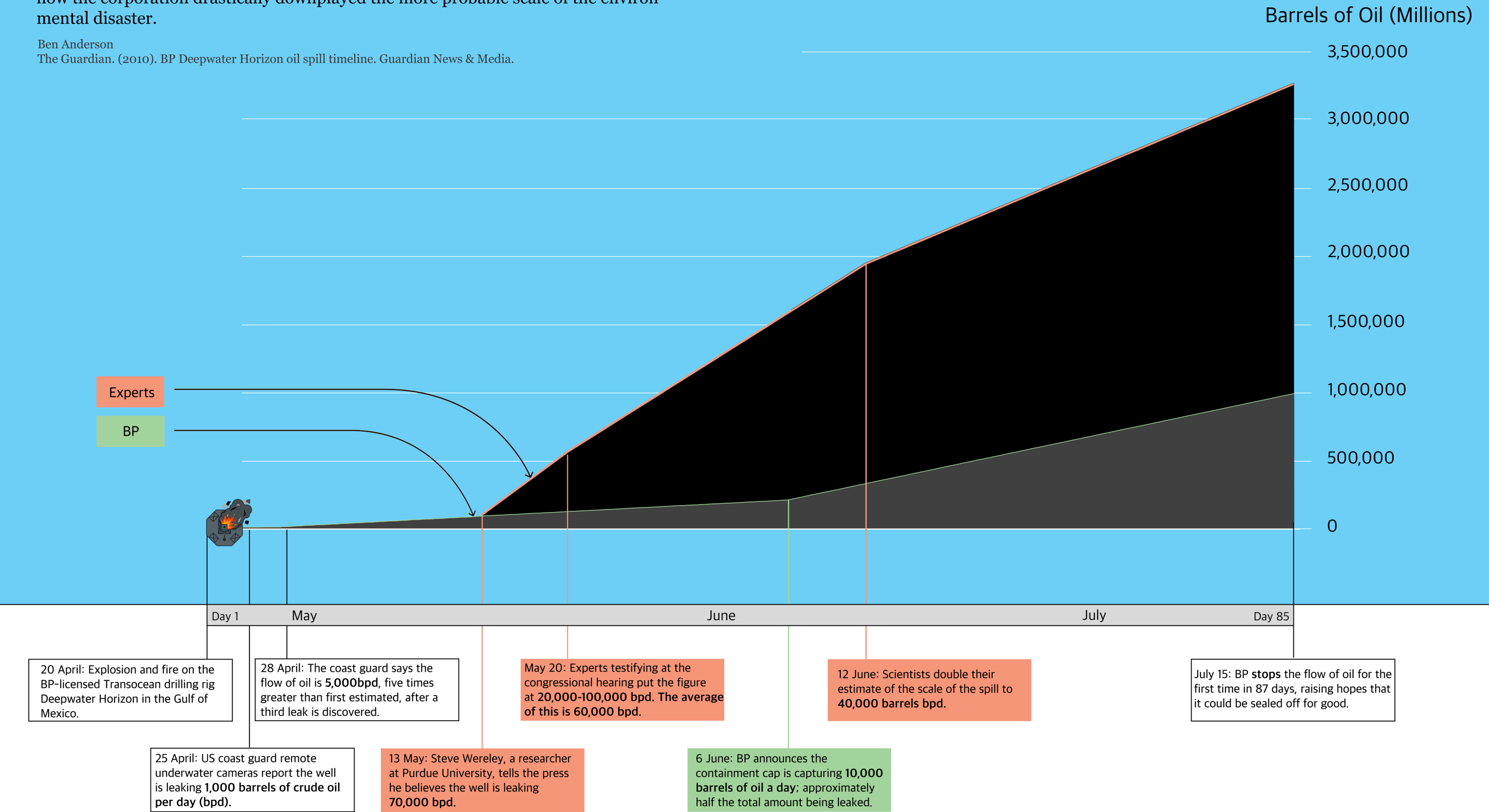


Timeline of the Cumulative BP Oil Spillage

This chart contrasts BP's minimized estimates of the 2010 Deepwater Horizon oil spill with the significantly larger estimates from independent experts. It highlights how the corporation drastically downplayed the more probable scale of the environmental disaster.

Ben Anderson
The Guardian. (2010). BP Deepwater Horizon oil spill timeline. Guardian News & Media.



Sequence Intent Statement

Problem

BP systematically underreported the scale of the Deepwater Horizon oil spill while independent experts estimated figures up to 20x higher. The gap between corporate reporting and scientific evidence was never clearly visualized in a single view.

Audience

General public, journalists, and students who need to understand the scale of corporate misrepresentation during the 2010 spill — without requiring technical background in oil extraction or environmental science.

Objective

Make the contrast between BP's estimates and independent expert estimates immediately legible, and show how that gap compounded over 85 days — turning a corporate communications decision into a quantifiable environmental accountability failure.

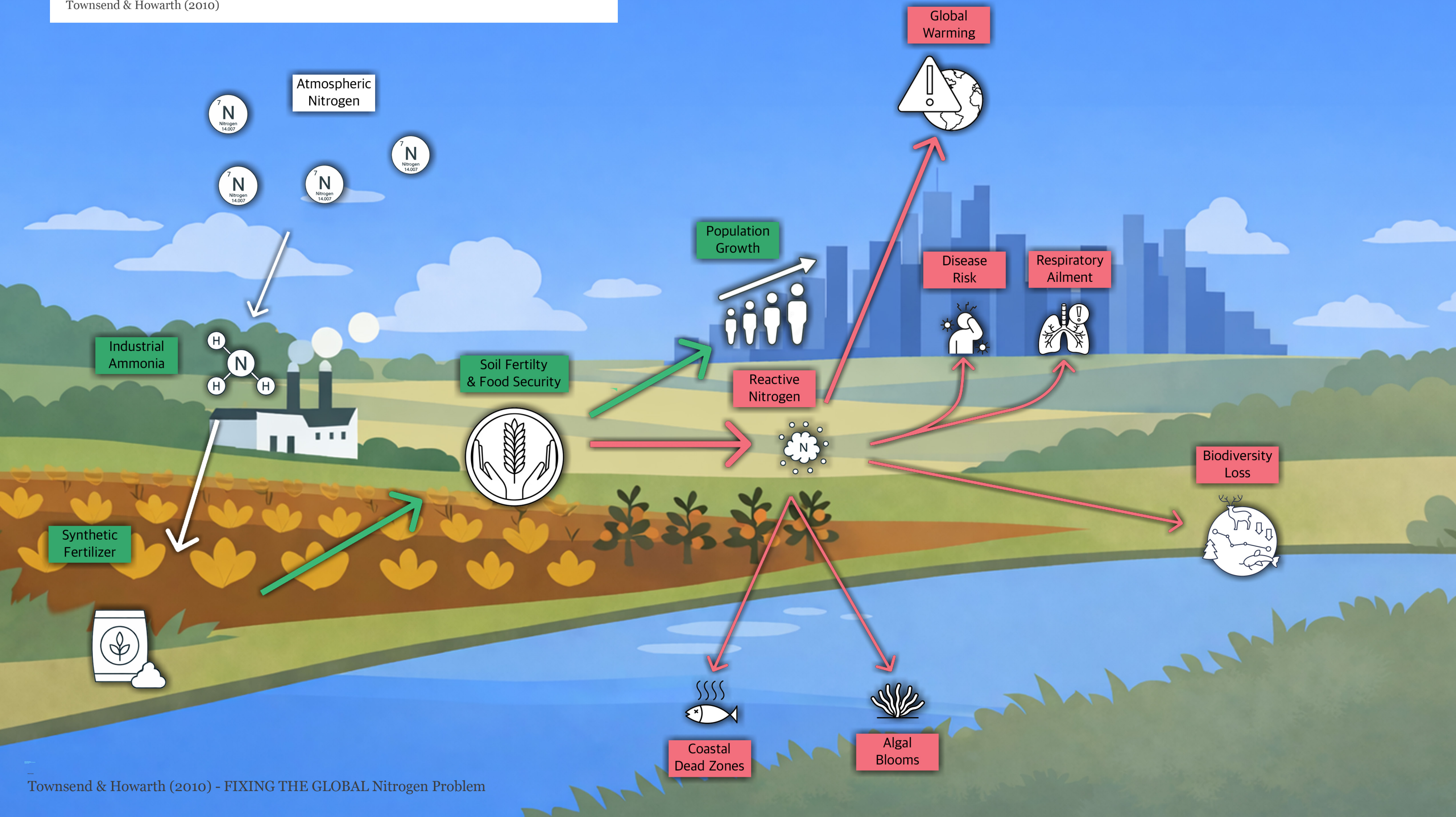
Approach

A cumulative area chart using two overlapping fills — BP's figures in dark grey beneath, expert estimates in black above — plotted against a shared timeline from Day 1 to Day 85. Key moments where estimates diverged are annotated directly on the chart. Color-coded event callouts distinguish BP statements from expert/government findings, making the pattern of underreporting traceable chronologically.

The Process of Nitrogen to Fertilizer

And its **intended/unintended** effects.

Ben Anderson
Townsend & Howarth (2010)



Process Intent Statement

Problem

The industrial nitrogen-to-fertilizer process was designed to solve food scarcity but generates reactive nitrogen as a byproduct — triggering a cascade of unintended environmental and public health consequences that are rarely connected back to their agricultural source in public understanding.

Audience

General public and students who understand fertilizer as a farming input but have no visibility into the downstream chain of consequences — from soil to waterway to atmosphere to human health.

Objective

Show that the same process producing food security simultaneously produces global warming, disease risk, respiratory ailment, biodiversity loss, coastal dead zones, and algal blooms — making visible the full cost of a system most people only see at the input stage.

Approach

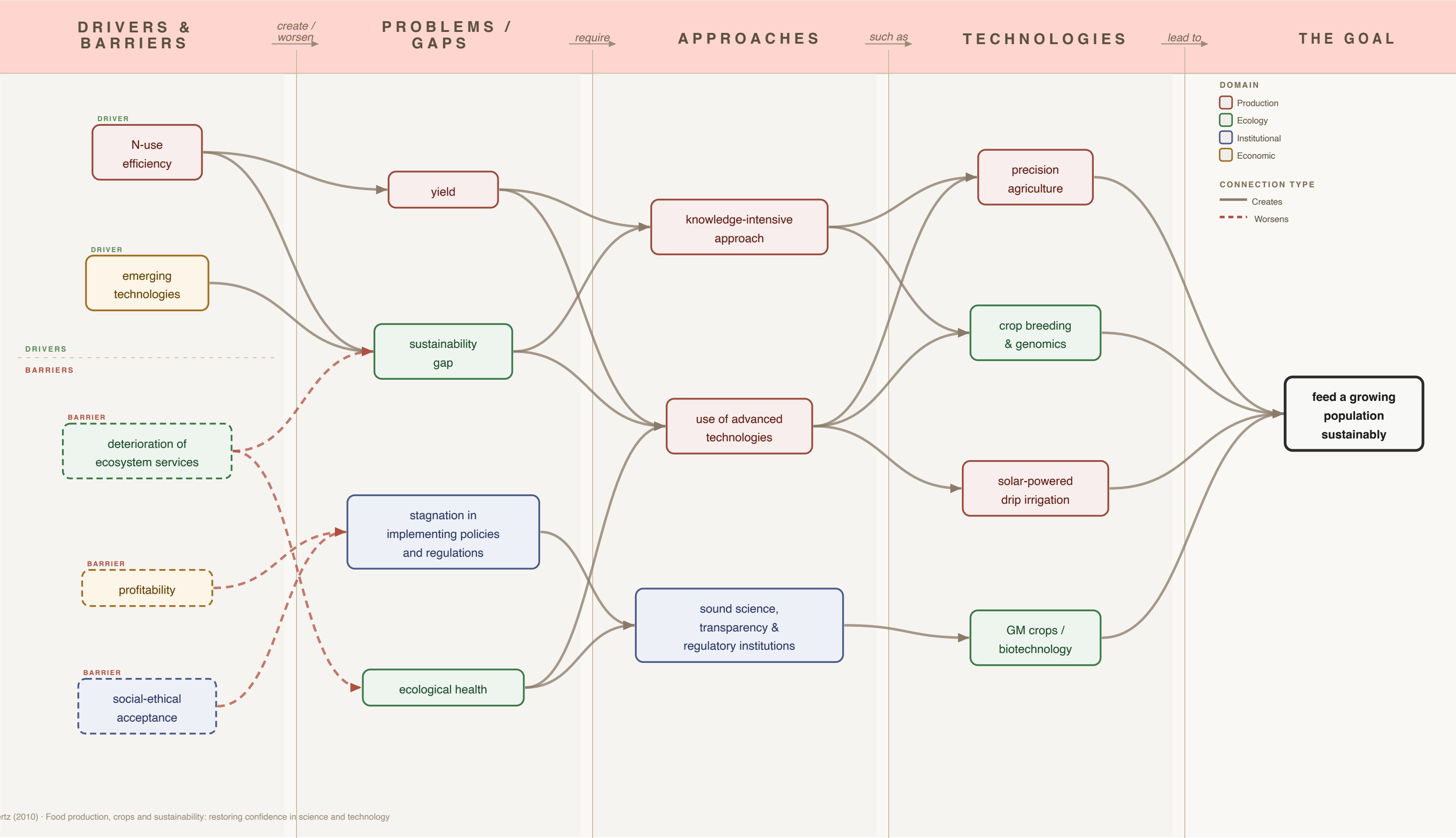
An illustrated landscape diagram that uses physical geography as the organizing logic — atmosphere above, land in the middle, water below — so the viewer can trace nitrogen's movement through environments intuitively. Green arrows mark the intended pathway, red arrows mark the unintended pathway caused by reactive nitrogen. The color contrast between green and red carries the intended/unintended distinction without requiring a legend.

Barriers to Solutions Within the Food Production Cycle

Ben Anderson · Spiertz (2010)

CONTEXT

By 2050, global population will reach 9-10 billion while cropland per capita shrinks from 0.25 to 0.15 ha/person. Food production must raise productivity and reduce environmental impact - yet current systems face compounding barriers.



Wildcard Connections Intent Statement

Problem

The academic literature on food production sustainability presents barriers, gaps, approaches, and technologies as separate discussions across multiple sections — making it difficult to see how they causally connect, which actors are responsible, and why progress remains slow despite available solutions.

Audience

Graduate students and researchers reading Spiertz (2010) who need to understand the paper's argument as a system rather than a linear sequence of points — and instructors evaluating whether a reader has genuinely understood the causal logic, not just the vocabulary.

Objective

Make the paper's implicit causal argument explicit in a single view — showing that specific drivers and barriers create specific gaps, which demand specific approaches, implemented through specific technologies, all oriented toward two quantitative targets the paper names directly.

Approach

A five-column left-to-right connection diagram using verbatim highlighted terms from the paper as nodes. Columns are separated by hairline dividers with conjunction labels describing the relationship between each stage. Node color encodes domain so the reader can identify the group each node belongs to. Line style distinguishes enabling connections from blocking ones. The first column splits drivers from barriers visually to reflect that not all inputs to the problem are the same kind of force. A context banner anchors the diagram in the paper's stated 2050 population and cropland figures before the logic begins.